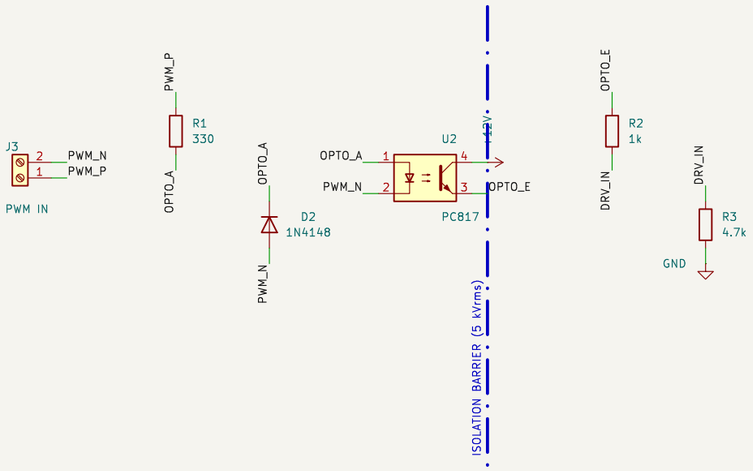
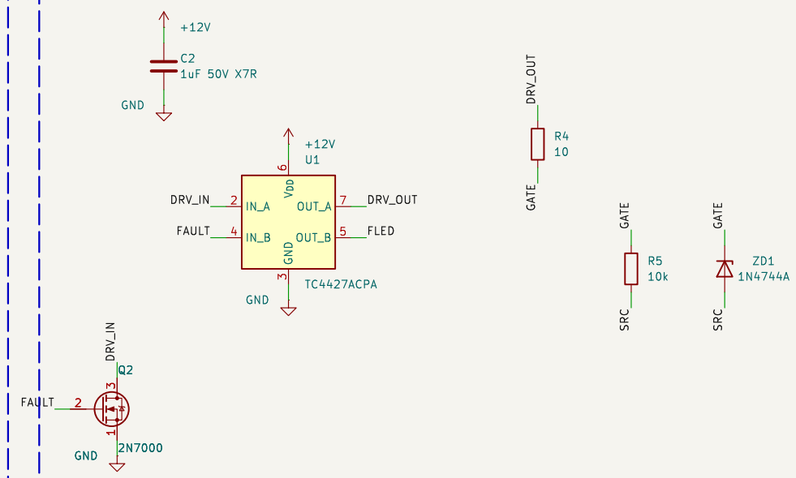
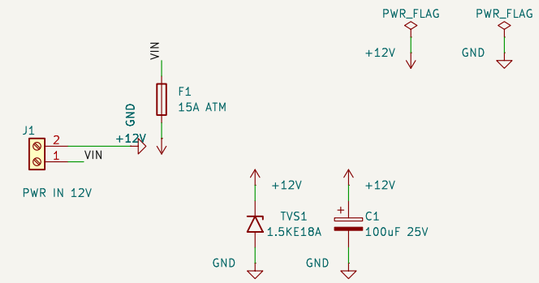
1 ISOLATED PWM INPUT (MCU side | power side)



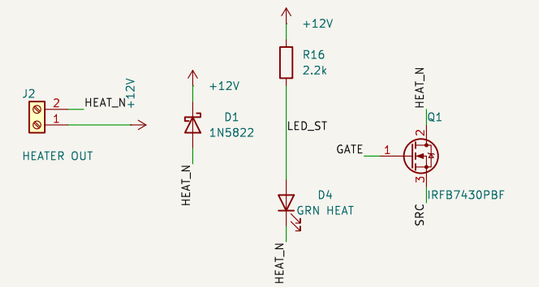
2 GATE DRIVE + FAULT SHUTDOWN



3 POWER INPUT + PROTECTION



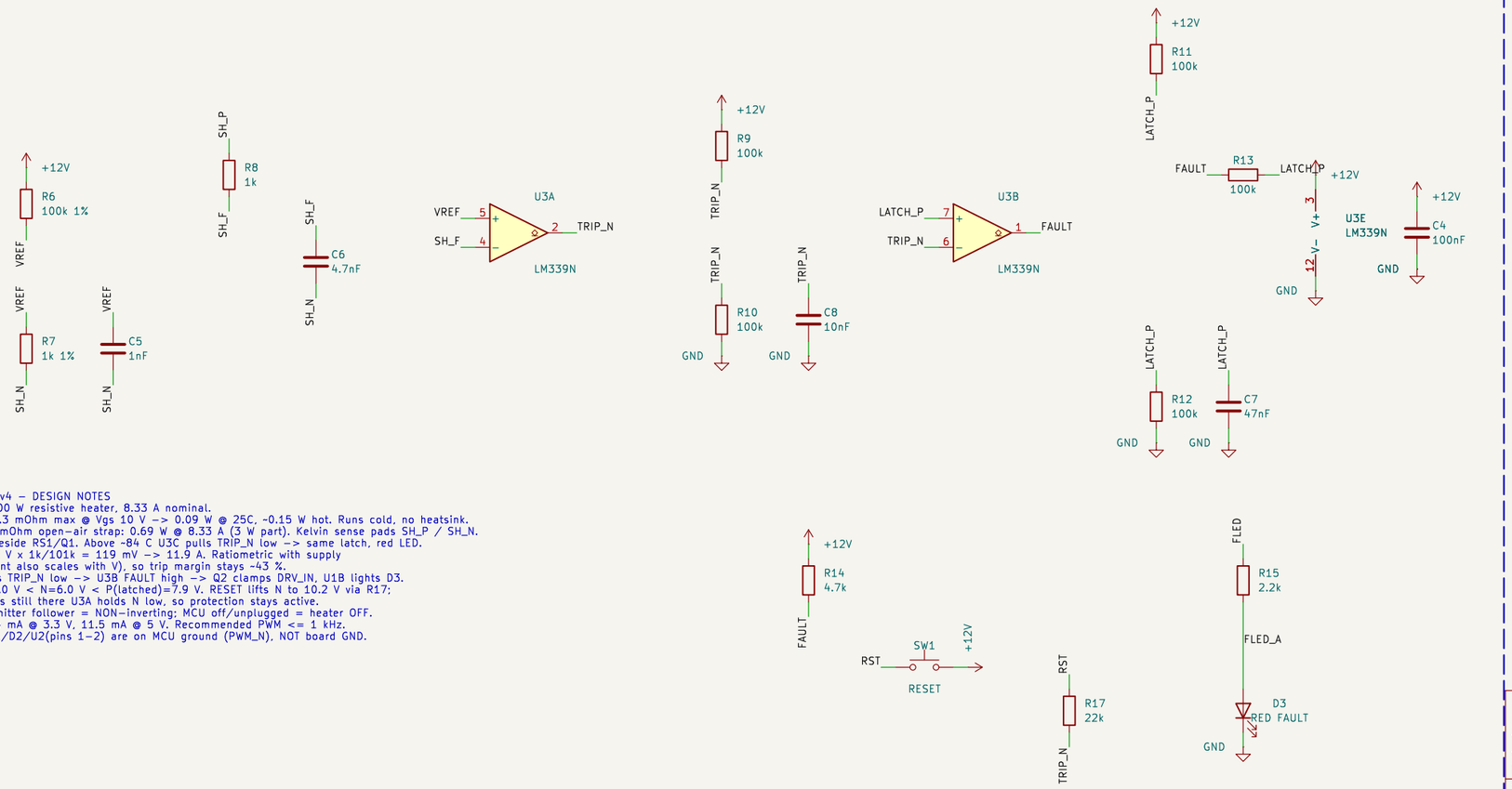
4 POWER STAGE (8.33 A)



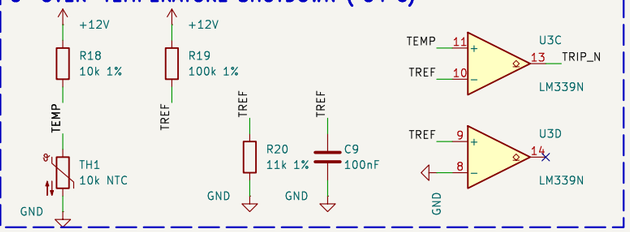
MECHANICAL



5 OVERCURRENT TRIP (11.9 A) + LATCH



6 OVER-TEMPERATURE SHUTDOWN (~84 C)



HEATER_DRIVER v4 — DESIGN NOTES
Load: 12 V / 100 W resistive heater, 8.33 A nominal.
Q1 IRFB7430: 1.3 mOhm max @ Vgs 10 V -> 0.09 W @ 25C, ~0.15 W hot. Runs cold, no heatsink.
RS1 OAR-3 10 mOhm open-air strap: 0.69 W @ 8.33 A (3 W part). Kelvin sense pads SH_P / SH_N.
OTP: TH1 NTC beside RS1/Q1. Above ~84 C U3C pulls TRIP_N low -> same latch, red LED.
Trip: VREF = 12 V x 1k/101k = 119 mV -> 11.9 A. Ratiometric with supply
(heater current also scales with V), so trip margin stays ~43 %.
Latch: U3A pulls TRIP_N low -> U3B FAULT high -> Q2 clamps DRV_IN, U1B lights D3.
P(normal)=4.0 V < N=6.0 V < P(latched)=7.9 V. RESET lifts N to 10.2 V via R17;
if the short is still there U3A holds N low, so protection stays active.
Input: PC817 emitter follower = NON-inverting; MCU off/unplugged = heater OFF.
R1 330R: 6.4 mA @ 3.3 V, 11.5 mA @ 5 V. Recommended PWM <= 1 kHz.
Isolation: J3/R1/D2/U2(pins 1-2) are on MCU ground (PWM_N), NOT board GND.

v4: 100 % through-hole parts. 2-layer 1 oz. Over-temperature + overcurrent latch.
v4: IRFB7430 1.3 mOhm, OAR-3 THT Kelvin shunt, LM339 OCP+OTP, safe RESET

MosfetHeater

Sheet: /
File: heater_driver_v4.kicad_sch

Title: Heater Driver 12 V / 100 W — v4 (All-THT, Isolated, OCP + OTP Latch

Size: A3	Date: 2026-10-17	**Rev: 4.0**
KiCad E.D.A. 9.0.9		Id: 1/1